

A topology-first and demand-driven sensor placement framework for hydraulic-state reconstruction in urban drainage networks

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HIGHLIGHTS

- An end-to-end framework enabled sparse-sensing inference for risk monitoring.
- An accurate, physically interpretable method proposed for sensor network assessment.
- The topology-to-utility metric quantified a priori gains for sensor placement.
- Supports demand-driven deployment and sensor-number determination.

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ABSTRACT

Monitoring data from urban drainage sensor networks are fundamental for system-state perception and risk-management decision-making. However, existing sensor placement (SP) studies primarily focus on detection accuracy for specific monitoring tasks while overlooking the contribution of SP to system-wide perceptibility, and typically rely on full-node hydraulic state data that is rarely available in practice. To address these limitations, we proposed a topology-first and demand-driven end-to-end framework spanning sensor placement, perceptibility assessment, and monitoring applications. The framework introduces the ideal perceptual domain (IPD) to quantify the theoretical perceptibility of sensor configurations and guide demand-driven placement optimization, while a graph-based reconstruction model is used to estimate the actual perceptual domain (APD) and support monitoring tasks. In the case study, the sensor network designed to achieve a target IPD coverage of 90% required sensors at only 3.0% of the network nodes. Under a representative rainfall event, this configuration achieved an APD coverage of 86.7%, an IPD–APD consistency coefficient of 0.96, and flooding-node identification precision and recall values of 0.95 and 0.91, respectively. The proposed framework provides a practical and interpretable solution with potential engineering applications in sensor-network design for urban drainage systems under data-limited conditions, thereby supporting urban flood-risk monitoring and decision-making for drainage-infrastructure management.

1. Introduction

Urban waterlogging, exacerbated by global climate change and urbanization, poses rising threats to public safety, economies, and infrastructure (Wu et al., 2025). Addressing this challenge requires real-time, system-wide monitoring of urban drainage networks (UDNs) to shift from reactive to proactive management. Water level serves as a critical indicator of drainage capacity and overflow risk, enabling assessment of flood losses and public safety threats. While sensor deployment at all

nodes would offer complete visibility and early anomaly detection, the cost and operational complexity of such dense networks are often prohibitive. Therefore, optimizing sensor placement is essential for balancing monitoring cost and system-wide observability which enables real-time assessment of network-wide operational states with limited sensing resources.

From a mathematical perspective, sensor placement can be formulated as a combinatorial optimization problem based on one or more objective functions, involving three fundamental questions: (1)

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determining the size of the candidate solution space, (2) identifying the criteria for optimality, and (3) selecting the most effective combination from a potentially large number of alternatives. Effectively identifying the optimal or near-optimal sensor layout from a solution space that grows exponentially with network complexity presents a significant challenge.

Optimization algorithms commonly used in existing studies, including heuristic and clustering algorithms, etc. (Kim et al., 2022; Marlim & Kang, 2021; Wang et al., 2023; Winter et al., 2019), typically rely on full-node hydraulic state data to acquire information for designing sensor networks across various monitoring tasks, such as stormwater systems (Wang et al., 2018; Wu et al., 2025), water distribution systems (WDNs) (Khorshidi et al., 2018, 2020; Tornyeiadzi et al., 2024), and sewer systems (Crowley et al., 2025; Wang et al., 2023; Wang et al., 2024). However, this assumption is impractical in real-world scenarios, where acquiring system-wide, node-level observation data is costly and technically challenging (Li et al., 2024).

To alleviate the reliance on complete system-wide data and enable sensor placement design under data-limited conditions, recent studies have begun to explore topology-driven optimization frameworks. Giudicianni et al. (2020) proposed a sensor placement method for contamination events warning in WDNs that is entirely based on topological information. In this approach, the WDNs are modeled as an undirected weighted graph. The Girvan–Newman clustering algorithm is used to partition the network based on edge betweenness. Five topological centrality metrics are then employed to assess node importance within each cluster, and the node with the highest centrality in each cluster is selected as the sensor location. Cheng & Li (2023) employed the graph localization operator to represent the diffusion characteristics of user nodes in WDNs and compute their importance and redundancy for sensor placement, thereby eliminating the need for hydraulic models in the node selection process. Although effective for sensor placement in WDNs, these methods rely on undirected network representations that are ill-suited to the directional and flow-driven nature of UDNs. Another important limitation of existing sensor placement strategies is that the number of sensors must be specified a priori, preventing a quantitative characterization of how sensing performance scales with sensor density. As a result, additional sensors might be deployed with limited gains in monitoring performance. Therefore, for urban drainage systems, there is a need to develop topology-driven sensor placement strategies, so that sensor network scale can be determined in accordance with target monitoring performance.

The primary purpose of sensor deployment is to monitor the operational status of UDNs, providing basic data support for flood monitoring and subsequent tasks such as flood-risk forecasting. Therefore, the ability of a sensor network to perceive the states of unmonitored nodes is regarded as a key aspect of performance evaluation (Zheng et al., 2025). This issue has attracted increasing attention in related fields. For example, Taha et al. (2021) and Bartos & Kerkez (2021) developed model-based approaches using Kalman filtering and observability-Gramian analysis, respectively, to optimize sensor locations under prescribed dynamic models. More recently, Oh & Bartos, (2025) proposed a QR-based framework for streamflow monitoring-network design and evaluated sensor layouts through reconstruction at ungauged locations. Zheng et al. (2025) explored the use of matrix completion to evaluate the reconstruction-oriented perceptual capability of optimized sensor layouts in UDNs, although the resulting reconstruction performance remained limited. Despite these advances, existing approaches typically rely on explicit dynamic models or complete system-wide observations for training. Therefore, an accurate and interpretable method is still needed to evaluate the system-wide perceptibility of sensor networks in data-limited UDNs.

Graph neural networks (GNNs) are a class of deep learning algorithms designed for graph-structured data. By leveraging message-passing frameworks that propagate, aggregate, and update information across connected nodes, GNNs show great potential for capturing

the spatial and physical dynamics of UDNs. For example, GNNs have been widely applied in the domain of UDNs, including water level estimation, reconstruction of missing topological attributes, surrogate modeling, etc (Garzón et al., 2024; He, Nan, Chen, et al., 2025a; Zhang et al., 2024).

Inspired by the powerful capability of GNNs in processing non-Euclidean structured data, we have found that the topological structure of UDNs can be exploited to propagate monitoring information along topological connections to neighboring nodes, enabling the reconstruction of unmonitored data through the integration of physical information. Theoretically, a well-designed sensor layout allows the reconstruction of system-wide unmonitored node data from limited monitoring nodes, enabling comprehensive awareness of UDNs operations and facilitating the effective monitoring of urban flooding and related disaster risks.

Building on these foundations, this study proposes an end-to-end framework that integrates topology-driven, demand-oriented sensor placement with GNN-based system-wide state reconstruction. The main objectives of this research are as follows: (1) provide a theoretical and methodological basis for designing sensor networks using only network topology in practically data-limited systems, (2) develop a more accurate and interpretable method for evaluating the performance of sensor networks, and (3) enable a paradigm shift in sensor deployment from preset-number-based design to demand-driven configuration based on system perceptibility.

2. Methodology

The workflow of the proposed framework is illustrated in Fig. 1. The framework first constructs the ideal perceptibility domain (IPD) of each node based on the network topology, which characterizes the theoretical observability of unmonitored nodes from candidate sensor locations. The target system-wide IPD coverage is then adopted as the optimization objective, and sensor locations are determined by seeking the minimum number of sensors required to achieve the specified IPD target. The optimized sensor configuration is then treated as the set of observation nodes and fed into a GNN-based reconstruction model. The model leverages the message-passing mechanism of GNNs to infer the operational states of unmonitored nodes. The reconstruction accuracy at unmonitored nodes is further adopted as the core metric for evaluating the performance of the sensor network, directly quantifying its ability to support system state reconstruction and inference tasks. Finally, the reconstructed system-level state information supports a variety of monitoring and risk identification tasks, forming a closed-loop framework that tightly couples sensor placement optimization, state reconstruction, and performance evaluation.

2.1. Sensor placement optimization

2.1.1. Defining the Optimization Objective

Based on the message-passing capability of GNNs, our preliminary experiments (see the supplementary material Part A) indicate that monitoring data collected from deployed sensors in UDNs can be used to infer the states of neighboring unmonitored nodes, with the inference range expanding as the number of propagation layers increases. The information propagation between nodes is influenced by the physical attributes of their connecting paths, such as pipe diameter, pipe length, and invert elevation. In other words, given the water level at a particular node and the physical properties of its connected pipes, the water levels at other nodes can theoretically be inferred through information propagation along the network topology. It should be noted that the physical edge attributes of the network topology, such as pipe diameter, pipe length, and invert elevation, are used to construct the Edge-STGCN model but are not incorporated into the subsequent sensor-placement stage.

Based on this, the sensor placement problem is formulated as a multi-

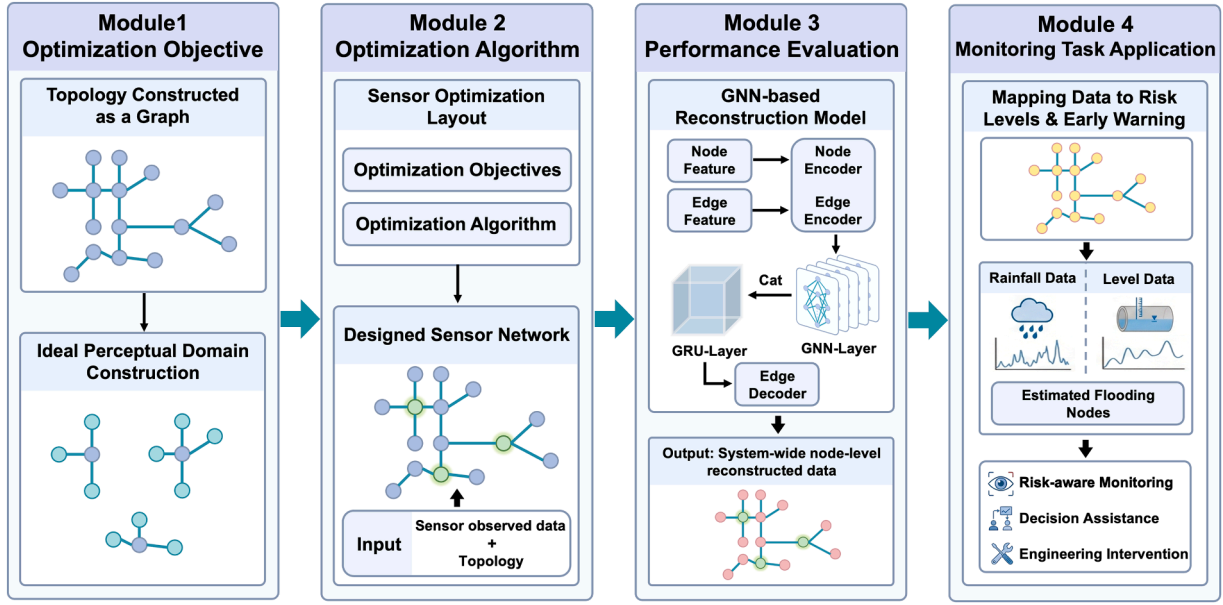


Fig. 1. The workflow of the proposed sensor placement optimization framework.

objective optimization task, underpinned by the message-passing mechanisms of GNNs. Sensor placement is optimized with two objectives: (1) **minimizing deployment costs** and (2) **maximizing IPD of the selected sensors**.

Definition 1. In this study, the cost of sensor deployment is quantified based on the number of sensors installed. However, in practical engineering applications, deployment cost is also influenced by additional factors such as the location-specific constraints and installation complexity associated with each sensor site, which are not within the scope of this study. This simplified treatment is also consistent with the approach commonly adopted in related sensor-placement studies (Wang et al., 2024; Zheng et al., 2025). For future engineering applications, infeasible candidate nodes, such as those subject to structural, spatial, or administrative constraints, can first be removed from the search space. The remaining feasible nodes can then be assigned location-specific cost coefficients according to their installation, communication, maintenance, and management requirements. These individual cost components can be integrated into a composite cost for each candidate node, after which the sensor-placement problem can be reformulated to minimize the total weighted deployment cost while satisfying the prescribed perceptual-coverage requirement. The sensor subset $\mathcal{J}$ is defined as Eq. (1).

$$\mathcal{J} = \{i^* \in V\} \subseteq V, |\mathcal{J}| = r \quad (1)$$

where r denotes the number of selected sensors, i^* denotes a selected sensor.

Definition 2. For a given directed graph $\mathcal{G} = (V, E)$, the k -hop IPD $\mathcal{N}^k(i)$ of a node $i \in V$ is defined as the union of its directed upstream and downstream neighborhoods within at most k hops. The upstream and downstream domains are constructed separately by recursively tracing predecessor nodes along incoming-flow paths and successor nodes along outgoing-flow paths, respectively. In this study, IPD serves as a direction-aware topological surrogate metric representing potential information propagation within the network. The relevant definitions and computational formulations are presented as follows.

$$\mathcal{N}^0(i) = \{i\} \quad (2)$$

$$\Gamma(u) = \Gamma^+(u) \cup \Gamma^-(u) \quad (3)$$

$$\mathcal{N}^k(i) = \mathcal{N}^{k-1}(i) \cup \bigcup_{u \in \mathcal{N}^{k-1}(i)} \Gamma(u) \quad (4)$$

where $\Gamma^+(u)$ and $\Gamma^-(u)$ denote the sets of downstream and upstream nodes directly connected to node u , respectively, $\mathcal{N}^0(i)$ is the set of neighboring nodes of node i at search depth $k = 0$, which by definition contains only the node i itself. The k -hop depth is set to 10 in this study, and the corresponding sensitivity analysis and parameter selection rationale are provided in the supplementary material Part B.

Definition 3. The overall IPD achieved by a sensor subset $\mathcal{J}$ is defined in Eq. (5):

$$\mathcal{E}(\mathcal{J}) = \bigcup_{i \in \mathcal{J}} \mathcal{N}^k(i) \quad (5)$$

The goal is to identify the minimal subset $\mathcal{J}^* \subseteq V$ such that the IPD $\mathcal{E}(\mathcal{J}^*)$ satisfies a user-defined coverage threshold $\gamma \in (0, 1)$, i.e.,

$$\frac{|\mathcal{E}(\mathcal{J}^*)|}{|V|} \geq \gamma \quad (6)$$

Where $|\mathcal{E}(\mathcal{J}^*)|$ denotes the number of IPD-covered nodes, and $|V|$ denotes the total number of nodes in the network.

An illustrative example of the IPD computation for individual nodes is provided in the supplemental material Part C.

2.1.2. Proposed optimization strategy

Sensors are selected greedily according to their marginal contribution to IPD coverage. After each selection, covered regions are removed from the remaining candidates to reduce redundancy. The process continues until the target coverage is reached, implicitly determining the required sensor number and yielding a deterministic sensor placement solution. The pseudocode of proposed strategy for sensor deployment is provided in Table S1.

2.1.3. Alternative optimization approaches for benchmarking

Two groups of comparative experiments are designed to validate the methodological advantages of the proposed framework and the rationality of its objective function, as described in the following subsections.

The first group of experiments is designed to validate the effectiveness and superiority of using the IPD as an optimization objective. Under the same number of deployed sensors, the proposed method is compared with two representative non-hydraulic-simulation-based methods re-

ported in the literature. The first method is a heuristic feature-selection algorithm. Cheng & Li, (2023) employed graph signal processing (GSP) to identify nodes that provide informative and non-redundant signals. Using both the amount of effective information contributed by each candidate node and the redundancy between the candidate node and the already-selected nodes as optimization objectives, they formulated a greedy optimization model to iteratively select sensor locations. This approach aims to identify nodes that maximize informational value while minimizing redundancy for sensor placement. As the related definitions and computational formulations are extensive and space is limited, interested readers are referred to the aforementioned literature for detailed descriptions. The second method is a topology-driven multi-objective genetic algorithm (TBNSGA-II). Wang et al. (2024), formulated sensor placement as a multi-objective optimization problem with sensing cost, sensing coverage, and expected search cost as the objective functions. They employed the NSGA-II model to derive a set of Pareto-optimal sensor layouts that maximize upstream coverage, minimize expected search cost, and reduce the number of sensors required. The relevant definitions and computational formulations are presented as follows.

$$\mathcal{S} = \sum_{i=1}^N \tau_i \quad (7)$$

$$\psi = \sum_i p_i \cdot O_i \quad (8)$$

$$\mathcal{F} = \sum_{i=1}^{\mathcal{S}} \frac{p_i}{\psi} \log_2(p_i) \quad (9)$$

where $\mathcal{S}$ represents the total number of sensors placed in the UDN, N represents the number of nodes in the UDN, $\tau_i = 1$ if there is sensor placed in this manhole, else $\tau_i = 0$, ψ represents the coverage of designed sensor network, p_i is the number of upstream manholes covered by sensor i , $O_i = 1$ if there is no sensor placed in its downstream area, else $O_i = 0$, $\mathcal{F}$ represents the spatial resolution of designed sensor network as the expected number of manual tests required to identify the exact location of a detect event.

The second group of experiments is designed to validate the superiority of the greedy algorithm under the same optimization objective. Specifically, with the objectives of maximizing the IPD and minimizing deployment costs, the greedy algorithm is compared with representative baselines, including the genetic algorithm (GA) and integer linear programming (ILP) (Liu et al., 2008; Romero-Ben et al., 2023). Detailed settings and parameter configurations for the ILP, Greedy, and GA algorithms are provided in the supplementary material Part E.

2.2. Sensor network evaluation

2.2.1. Evaluation methods

In this study, the graph-based Edge-STGCN model proposed by He, et al. (2025b) is adopted to evaluate the perception capability of the designed sensor network. This model integrates spatiotemporal graph convolution with pipe attributes, enabling the capture of dynamic propagation features among drainage nodes based on spatiotemporal dependency modeling, thereby achieving high-accuracy reconstruction of hydraulic states at unmonitored nodes. Readers interested in a comprehensive understanding of Edge-STGCN may refer to the aforementioned paper (He, et al., 2025b). This study focuses on introducing the training strategy of the Edge-STGCN model applied to sensor placement optimization, as detailed in the following subsection.

2.2.2. Training strategy of Edge-STGCN model

The proposed sensor placement strategy uses IPD coverage as the optimization objective. IPD provides a topology-based approximation of

potential information propagation between monitored and unmonitored nodes. Therefore, the optimization aims to reduce redundant overlap among IPDs and maximize the marginal coverage contribution of each sensor, thereby minimizing the number of sensors required to achieve a prescribed IPD target. This principle should be interpreted as a cost-saving heuristic rather than a general optimality condition. During the training of the Edge-STGCN model, the input is the monitoring data from sensor nodes, while the output is the reconstructed states of all nodes across the system. Relying solely on the permanently deployed sensors selected by the optimization strategy may result, under sparse sensor layouts, in a lack of supervisory signals from unmonitored areas within the corresponding IPDs in the training and validation data. Consequently, the Edge-STGCN model may receive insufficient guidance to effectively learn the reconstruction of states at unmonitored nodes. Therefore, during the training stage of the reconstruction model, a small number of unmonitored nodes are selected as temporarily deployed sensors to provide supervisory signals and compute reconstruction loss. Since this study aims to validate the effectiveness of the proposed framework while minimizing the influence of field-device errors, missing data, and operational uncertainties, water-level time series generated by the SWMM model are used to represent the observations at these temporary nodes instead of relying on field sensor measurements.

The temporarily deployed sensors are randomly selected from the feasible candidate nodes while maintaining an approximately uniform spatial distribution to avoid excessive concentration in local regions that could bias model training. Once determined, the set of temporarily deployed sensors remains unchanged across all model training, validation, and algorithm comparison experiments to ensure the comparability of experimental results. It should be noted that these temporarily deployed sensors are used only for the initial training and performance validation of the reconstruction model and can be interpreted as short-term water-level loggers deployed during the model calibration stage in practical engineering applications. During the operational reconstruction stage, the temporary sensors are removed, while the permanent sensors selected using the proposed placement strategy continue to provide the real-time observations required for system-wide state reconstruction. Considering that temporary sensors can be reused across different monitoring campaigns in practical applications, their amortized deployment cost may be substantially lower than that of permanently installed sensors. Therefore, temporary sensors were not included in the final deployment-cost metric in this study, which accounted only for the permanent sensors. Ten temporarily deployed sensors were used in this study.

During model training, a passive masking strategy is employed to enhance the model's generalization ability under sparse monitoring conditions, as shown in Fig. 2. Specifically, the initial input node state matrix retains observation values only at sensor-equipped nodes, while unmonitored nodes are assigned zero values to represent missing data. In each training batch, to prevent over-reliance on specific sensor inputs, 50% of the sensor nodes are randomly selected for masking, and their state features are replaced with zeros. Despite the masking operation during the forward pass, the loss function is computed over all sensor nodes. This ensures the model learns meaningful structural relationships and temporal patterns by leveraging neighborhood information and edge attributes, rather than memorizing sensor-specific features. To account for stochasticity, five independent repetitions were conducted using predefined random seeds. Within each repetition, all sensor-placement strategies used the same data split and model-training seed, enabling paired comparisons under identical Edge-STGCN initialization conditions. For deterministic placement methods, the sensor layouts remained unchanged under fixed inputs and parameter settings, and only the Edge-STGCN training process was repeated. For stochastic methods, including IPD-GA and TBNSGA-II, the complete placement-training-evaluation procedure was independently repeated, with each generated layout evaluated using the correspondingly seeded reconstruction model. A common seed selected based on validation-set

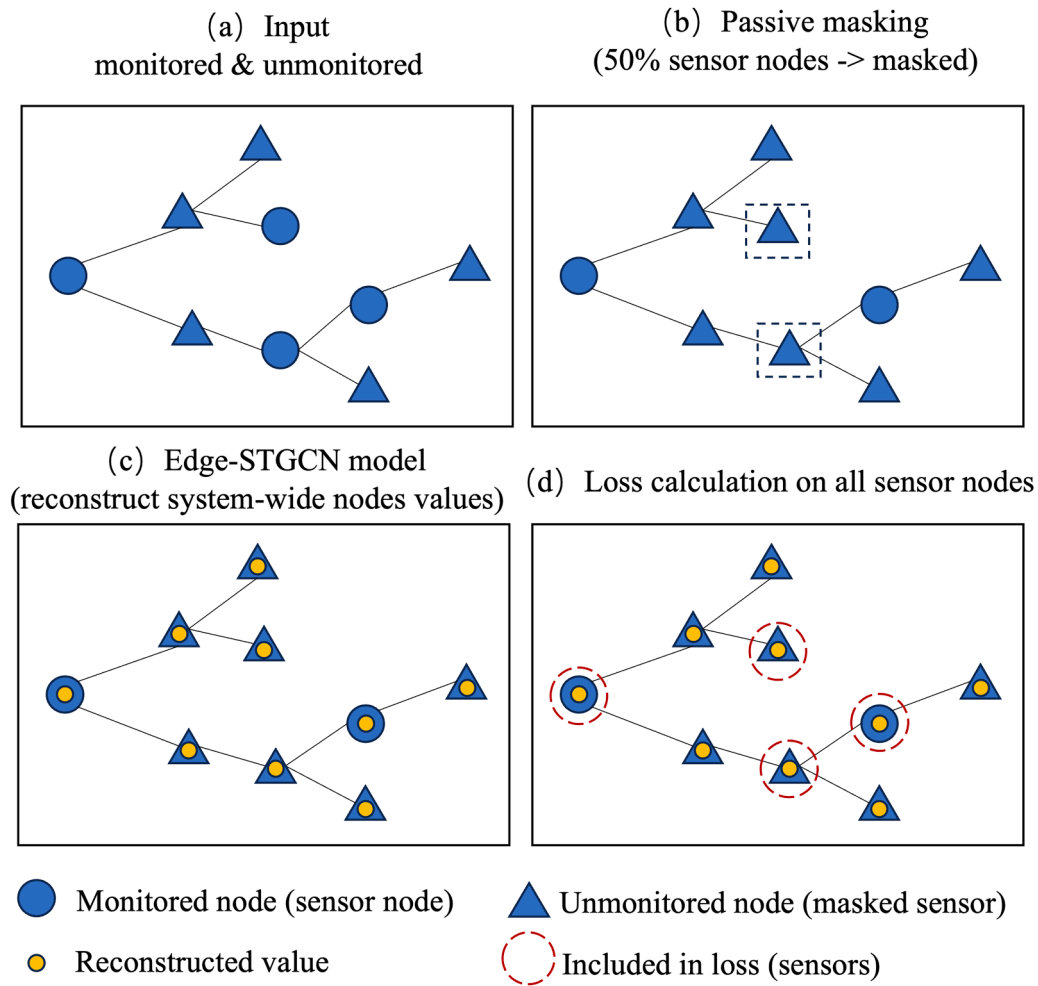


Fig. 2. The schematic diagram of passive masking strategy.

reconstruction performance was used for the representative spatial analyses in the main text. The results obtained using the remaining seeds, together with detailed information on the random seeds, are provided in the supplementary material (Part E and Part I).

2.2.3. Loss function of Edge-STGCN model

In the training process of Edge-STGCN model, the mean-squared error (MSE) and the hydraulic boundary-constraint loss are combined to form the overall training loss. The MSE term measures the average squared deviation between the true and reconstructed values of a specified node, providing a direct assessment of reconstruction accuracy in the data space. The formula is expressed in Eq. (10).

$$Loss_{mse} = \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i - \hat{x}_i)^2 \quad (10)$$

where N_s is the number of sensors selected for the loss calculation, x_i is the true water-level value, $\hat{x}_i$ is the reconstructed water-level value.

The hydraulic boundary-constraint loss is designed to prevent physically infeasible reconstruction results. For the lower boundary, a penalty is applied when the reconstructed water level $\hat{x}$ fell below the invert elevation z_{min} , reflecting the physical constraint that nodal water levels should not be lower than the manhole bottom elevation. For the upper boundary, a conditional constraint is adopted. When the ground-truth value satisfies $x \leq z_{max}$, reconstructed values exceeding the ground elevation ($\hat{x} > z_{max}$) are penalized. Conversely, when $x > z_{max}$ indicating surcharge or overflow conditions, no upper-bound penalty is imposed. The proposed approach represents flooding at the node scale based on

the exceedance of one-dimensional hydraulic heads above ground elevation. The physics constraint loss penalizes violations of these bounds, defined as Eq. (11). The first term penalizes predictions falling below the pipe invert, while the second penalizes predictions exceeding the ground surface elevation. The total loss is thus expressed as Eq. (12).

$$Loss_{phy} = \frac{1}{n_s} \sum_{i=1}^{n_s} \{ \max(0, z_{min,i} - \hat{x}_i)^2 + \max(0, \hat{x}_i - z_{max,i})^2 | x_i \leq z_{max,i} \} \quad (11)$$

$$loss = loss_{mse} + \lambda loss_{phy} \quad (12)$$

where λ is a tunable hyperparameter that balances data fidelity and physical consistency, set to $\lambda = 1$ in this study.

2.2.4. Performance metrics

The model after training is used for evaluating the performance of the designed sensor network. The observations of sensor nodes are provided as input to the Edge-STGCN model and the outputs of reconstructed states for system-wide nodes are regarded as perception data by sensor network. Reconstruction accuracy is evaluated using the Nash–Sutcliffe efficiency (NSE), a metric widely used in hydraulic model calibration, and the percentage of nodes with $NSE \geq 0.5$ is used to quantify the actual perception domain (APD) of the designed sensor network, a threshold generally considered acceptable for engineering applications (Guo et al., 2023; Pang et al., 2020). It should be noted that, in this study, the APD is intended to serve as a general indicator of the sensor network's reconstruction capability. The calculation formulas Eq.

(13)-Eq. (14) are shown below.

$$NSE = 1 - \frac{\sum_{i=1}^n (\hat{H}_i - H_i)^2}{\sum_{i=1}^n (H_i - \bar{H}_i)^2} \quad (13)$$

$$P_{NSE \geq 0.5} = \frac{\sum_{i=1}^n f(NSE \geq 0.5)}{n} \quad (14)$$

In this study, mean absolute percentage error (MAPE) and root mean square error (RMSE), which are widely used in reconstruction accuracy assessment, are additionally adopted as complementary metrics to validate the reconstruction results. The calculation formulas Eq. (15)-Eq. (16) are shown below.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{H_i - \hat{H}_i}{H_i} \right| \quad (15)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{H}_i - H_i)^2} \quad (16)$$

where H_i is the true water-level value, $\hat{H}_i$ is the predicted water-level value, $\bar{H}_i$ is the mean of the true water-level values, and n is the number of system-wide nodes, $f(\cdot)$ is the indicator function, which equals 1 if the condition inside is true and 0 otherwise. The NSE value ranges from $(-\infty, 1]$, with $NSE = 1$ indicating a perfect fit between the reconstructed and true values.

In addition, an IPD-APD consistency metric η is introduced to quantify the agreement between theoretical perceptual coverage and actual state reconstruction capability. The η ranges from 0 to 1, with higher values indicating that IPD-based sensor placements are more effectively translated into actual reconstruction performance. Its calculation formula is as follows.

$$\eta = \frac{APD}{IPD} \quad (17)$$

In this study, perceptibility refers to the capability of a sensor network to provide sufficient information for reconstructing hydraulic states at unmonitored nodes. It is conceptually related to observability because both concern the inference of unmeasured system states from limited observations. However, perceptibility, as defined here, does not denote formal system-theoretic observability derived from a prescribed state-space model, an observability matrix, or an observability Gramian. The IPD represents an a priori, topology-derived proxy for the potential information-propagation range of a sensor configuration, whereas the APD is an a posteriori, reconstruction-based measure of the nodes whose hydraulic states can be reconstructed with acceptable performance.

2.3. Monitoring task application

The performance of flood risk assessment under storm events is evaluated using three standard classification metrics based on the states reconstructed from the designed sensor network (Wang et al., 2024).

Precision represents the percentage of correctly identified flooding nodes and total nodes identified as flooding node. Its calculation formula is as follows.

$$Precision = \frac{TP}{TP + FP} \quad (18)$$

Recall represents the percentage of correctly identified flooding nodes and total actual flooding nodes. Its calculation formula is as follows.

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

False positive rate (FPR) represents the percentage of non-flooding nodes that are incorrectly identified as flooding nodes. Its calculation formula is as follows.

$$FPR = \frac{FP}{FP + TN} \quad (20)$$

where TP denotes the number of nodes that are truly flooding and are correctly identified as flooding, TN denotes the number of nodes that are truly non-flooding and are correctly identified as non-flooding, FP denotes the number of nodes that are truly non-flooding but are incorrectly identified as flooding, FN denotes the number of nodes that are truly flooding but are incorrectly identified as non-flooding.

3. Case study

The case study is based on an urban drainage system in southeastern China, where river networks are densely distributed and the annual rainfall is approximately 1,609 mm. The system consists of 404 nodes (402 manholes, 1 pump station forebay, and 1 outfall) interconnected by 409 pipes, covering an area of 129 ha with a total pipe length of 10347 meters. It operates as a passive drainage system, with a pump station located at the downstream end, designed for a capacity of 10.72 m³/s. One existing sensor (E1) is installed in the area. E1 is treated as a fixed monitoring node and independent benchmark. It is excluded from the candidate sensor set during optimization and is used only at the final stage to provide an auxiliary field-based validation of the performance assessments produced by the different evaluation methods. Detailed network topology information is shown in Fig. 3.

The drainage network in the study area is constructed decades ago, and several pipelines suffer from undersized diameters and insufficient conveyance capacity. With rapid urbanization and the increasing frequency of extreme rainfall events, the area has repeatedly experienced urban flooding, revealing the system's pronounced vulnerability under high-intensity storm conditions. The local utility operators urgently require real-time and system-wide monitoring to gain a comprehensive understanding of network operation and to support scientific decision-making for flood prevention and emergency management. In light of these circumstances, this study selects the area as a case for developing and validating the proposed framework for sensor placement optimization, in order to assess its applicability and engineering value in typical aging drainage systems.

3.1. Data preparation

For the optimization of sensor placement, the required input consists solely of the UDN topology (as illustrated in Fig. 3), including network connectivity and flow-direction information. It should be noted that no physical edge attributes are used at this stage.

For training the reconstruction model, the required data include water-level observations from both permanent and temporary monitoring sensors, together with the UDN topology augmented with physical edge attributes. The effects of pump operation states and downstream boundary conditions are not considered as model inputs in this study, but are implicitly reflected in the observed water-level time series and captured through the learned spatiotemporal relationships among network nodes. Given that field-measured water levels are generally unavailable at unmonitored nodes throughout the network, the accuracy of the reconstructed states at these locations cannot be directly verified. Therefore, this study used a calibrated SWMM model provided by the utility operator and driven by real rainfall data to establish a controlled validation environment with a complete set of system-wide reference states. The full-node simulation outputs are used solely as post hoc reference values for performance evaluation and are not provided to the reconstruction model as field observations. Specifically, as the study area is served by a separated stormwater system, water-level event data are generated by dynamically simulating node water-level responses under multiple rainfall scenarios using the SWMM model. The rainfall inputs are generated using the alternating block

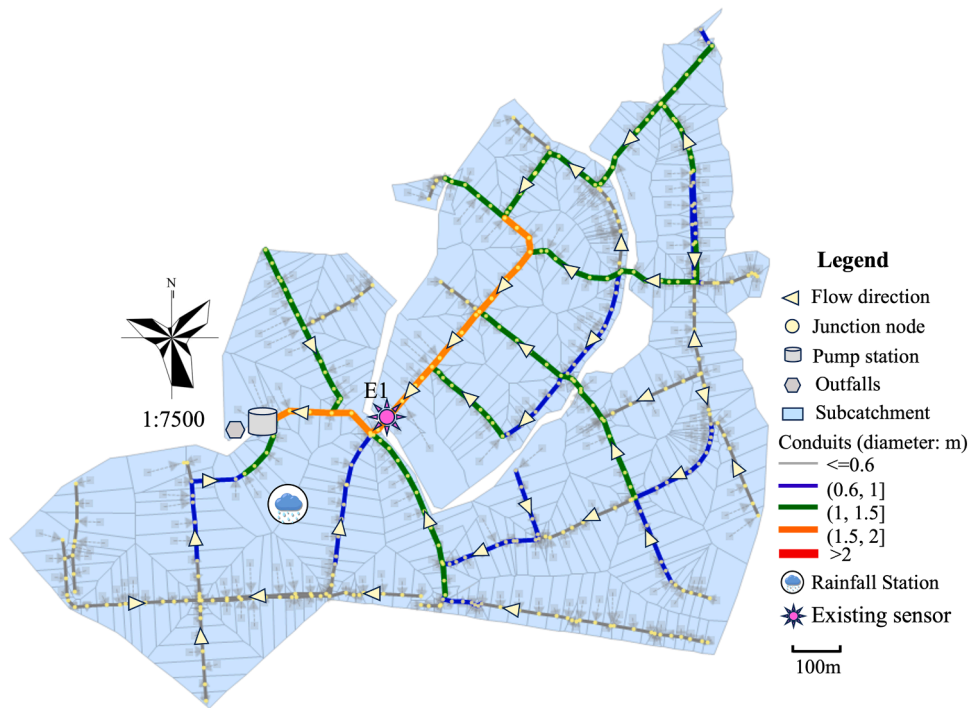


Fig. 3. The topological graph of study area.

method (Tobin, 1989) to produce synthetic rainfall, and the corresponding rainfall intensity and duration parameters are provided in the paper (Garzón et al., 2024). Previous studies have shown that incorporating historical observations from monitored nodes can improve hydraulic-state reconstruction at unmonitored nodes (He, Nan, Chen, et al., 2025a). Therefore, the input of the Edge-STGCN model is a sequence of node water-level matrices over 24 consecutive time steps, $\mathbb{X}_{t-23:t} \in \mathbb{R}^{n \times 24}$. The data from sensor nodes are provided as known observations, while those from unmonitored nodes are processed using a zero-mask scheme. The output of the model is the reconstructed water-level matrices of all nodes for the latter 12 time steps of the same input window, $\mathbb{Z}_{t-11:t} \in \mathbb{R}^{n \times 12}$. For example, for an input window spanning time steps 0-23, observations during time steps 0-11 provide historical hydraulic-evolution information, whereas those during time steps 12-23 provide current monitoring information. Based on these inputs, the model reconstructs the water levels of all nodes during time steps 12-23. During the training and validation process, only the monitoring data from the sensor-deployed nodes are used for parameter learning and model optimization.

For the operational reconstruction of unmonitored-node states, observations from the permanent sensor nodes selected through the proposed optimization procedure, excluding those from the temporary sensors, are fed into the trained model to reconstruct the water levels at unmonitored nodes. Finally, the perception capability of the sensor placement optimization scheme is evaluated by calculating the NSE between the observed and reconstructed values at all nodes.

The training and validation datasets are divided at the rainfall-event level with a ratio of 7:2. In addition, the proposed framework is also evaluated under multiple rainfall scenarios, including two real rainfall events and five design storms corresponding to 3, 5, 10, 50, and 100 year return periods. It should be noted that the term “real rainfall event” refers to an observed rainfall time series used to drive the simulated hydraulic response, rather than to a field-measured hydraulic response. Real rainfall event 1# on July 6, 2020 is selected as the representative case for testing and detailed analysis in the main text. Details of these remaining rainfall events and the corresponding validation analyses are provided in the supplementary material.

3.2. Technical implementation

All experiments were conducted on a GeForce RTX 4090 GPU, 16348CUDA 24 GB Turbo fan 2-slot, and AMD EPYC 7763/64Core/128Threads/2.45G/280W/256MB. Details regarding the hyper-parameter settings of the reconstruction model are provided in Table S3.

4. Results and discussion

4.1. Establishing a reliable perceptibility evaluation framework for sensor networks

A physics-enhanced reconstruction model was developed for sensor network performance evaluation, and its effectiveness was compared with that of a matrix completion method under conditions where full-node hydraulic state data were unavailable. Using a deployment scale of 10 sensors, a genetic algorithm was employed to optimize sensor placement with deployment cost, upstream coverage, and expected search cost as the objectives Wang et al. (2024). Based on the resulting sensor layout, the matrix completion method and the reconstruction model were applied under representative rainfall event 1# to evaluate sensor network performance.

It was observed that the reconstruction results of the matrix completion algorithm and the reconstruction model were able to restore the general variation pattern of the measured water-level at sensor E1, whereas clear differences in reconstruction accuracy were evident (Fig. 4(a)). The matrix completion algorithm yielded an NSE of 0.011, an RMSE of 0.670 m, and a MAPE of 69.690%, whereas the reconstruction model yielded an NSE of 0.927, an RMSE of 0.182 m, and a MAPE of 39.010%. Although RMSE values < 1 yielded by both approaches might have appeared acceptable, discrepancies were evident in the NSE metric, which was sensitive to peak water-level variations essential for urban flooding risk monitoring and early warning. This suggests that relying solely on RMSE may obscure substantial errors occurring during localized extreme events or abrupt state changes within the designed sensor network, potentially leading to failures in detecting or responding to sudden flooding incidents. In addition, both methods produced

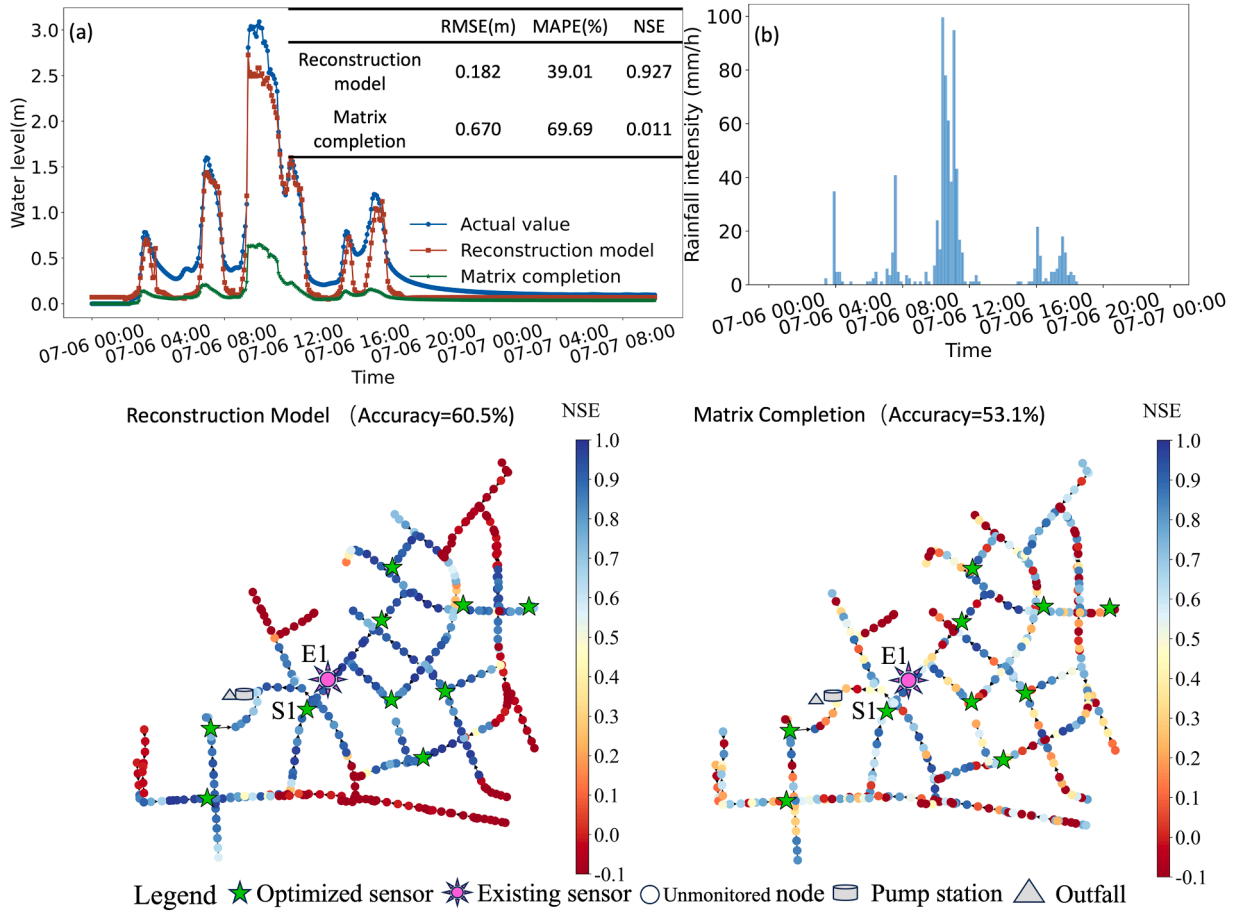


Fig. 4. (a) The comparison between the actual and reconstructed water levels (by matrix completion algorithm and reconstruction model) of the existing sensor E1, (b) Rainfall process (test data) in the study area on July 6, 2020, (c-d) The spatial distribution of system-wide node water-level reconstructed errors.

relatively high MAPE values because MAPE is highly sensitive to near-zero observed water levels and may inflate relative errors during low-water-level periods. Therefore, it was reasonable for this study to select the NSE metric as the main performance evaluation index for sensor networks. Unless otherwise specified, the subsequent reconstruction errors were evaluated using the NSE metric.

The spatial distribution of system-wide reconstruction errors for the designed sensor network was illustrated in Figs. 4(c) and 4(d). Although sensor E1 and sensor S1 were located in close spatial proximity, a substantial discrepancy was observed between the reconstructed values and the actual observations when employing matrix completion algorithm. Moreover, the reconstruction accuracy at unmonitored nodes showed no clear spatial correlation with the physical locations of the deployed sensors. In contrast, the reconstruction model exhibited a pronounced spatial dependency and achieved higher reconstruction accuracy. Fig. S4 further presented single-node perceptual domain experiments, which provided additional validation of the model's spatial dependency. The observed differences in spatial dependency were fundamentally attributed to the inherent distinctions between the two approaches in terms of their modeling mechanisms and information utilization strategies. Matrix completion was grounded in low-rank matrix approximation theory and typically searched for an optimal low-rank solution, starting from random initialization and constrained by the available observations, to reconstruct the missing data. The lack of topological information and its inherent randomness made the reconstruction outcome have no obvious regularity in spatial distribution. Consequently, its ability to distinguish the performance of different sensor layouts may be limited. This interpretation is further supported by the supplementary results (Part H) obtained by evaluating different sensor

layouts using the matrix-completion algorithm.

In contrast, the reconstruction model was built upon a GNN framework, where the topology and edge physical attributes were used for message passing and aggregation of monitoring node information to estimate the state of unmonitored nodes. The joint modeling of spatial connectivity and temporal dynamics enabled reconstruction results that were structurally consistent with the underlying network processes, thereby improving the stability and interpretability of system-wide state estimation. Therefore, the reconstruction model can be regarded as a more reliable approach for evaluating the perceptual performance of sensor networks.

4.2. Identifying a system-level optimization objective based on topological structure

Based on the findings of Section 4.1, a sensor placement strategy using the IPD as the optimization objective was developed as an intentional model-aware co-design between sensor placement and GNN-based reconstruction and was compared with existing topology-based sensor placement methods, including TBNSGA-II and GSP (Cheng & Li, 2023; Wang et al., 2024). As shown in Fig. 5(a), a clear divergence emerged in the reconstruction performance of the three sensor placement approaches as the number of sensors increased, with IPD-GA and GSP exhibiting consistently improving accuracy, whereas TBNSGA-II displayed instability once the number of sensors exceeded 12. For the same number of sensors, IPD-GA consistently outperformed both GSP and TBNSGA-II in terms of system-wide reconstruction accuracy. IPD-GA attained a maximum reconstruction accuracy of 92.1%, which is higher than that of GSP (83.0%) and TBNSGA-II (81.0%).

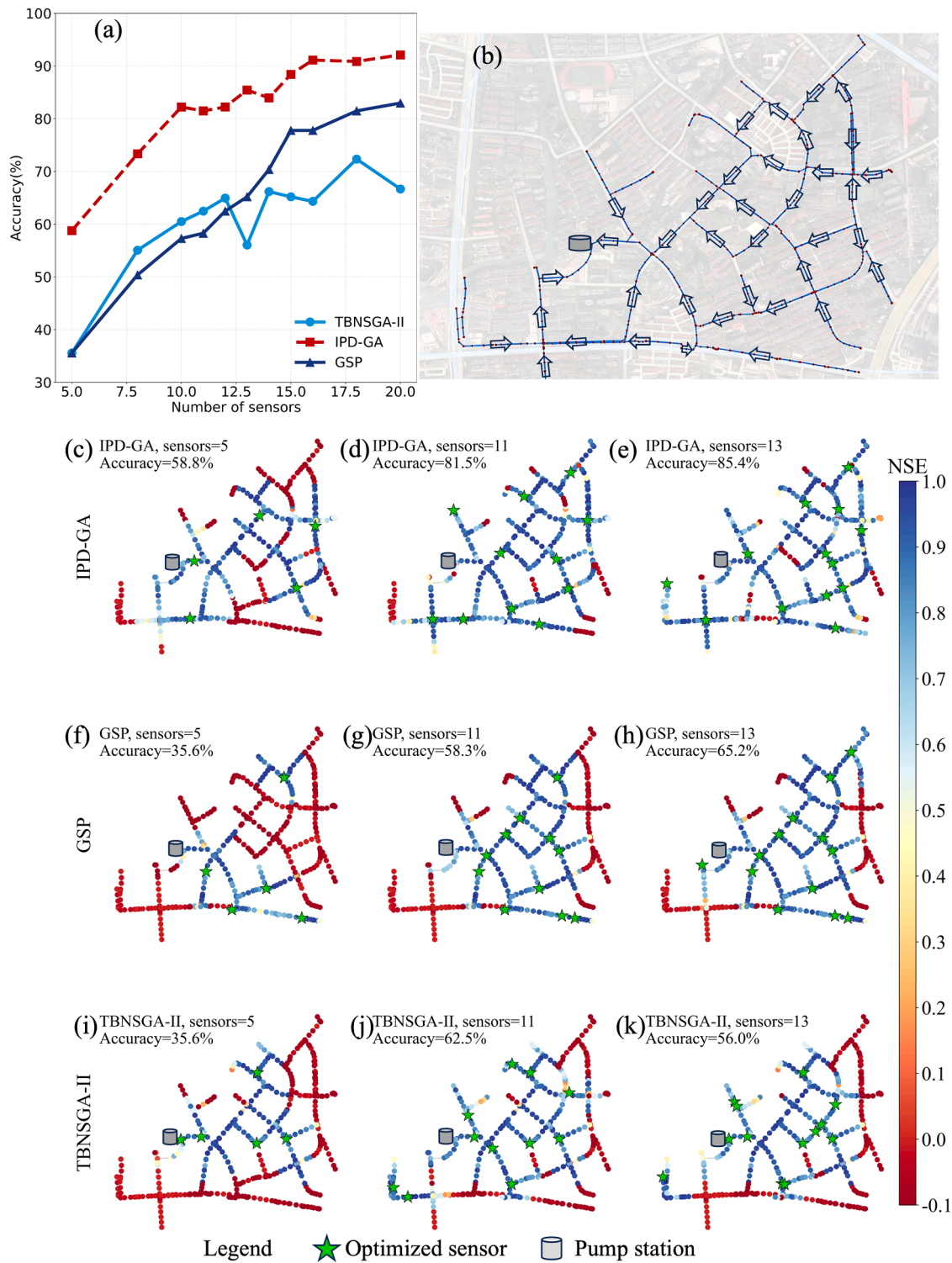


Fig. 5. (a) The reconstruction accuracy of UDNs under sensor placement strategies based on varying optimization objectives and sensor numbers. (b) Schematic diagram of the water flow direction in the study area. (c-k) The spatial distribution of system-wide node water-level reconstructed errors (by sensor placement strategies based on varying optimization objectives) under the specified number of sensors.

To further investigate the reasons for the observed differences in reconstruction performance, particularly the performance decline of the TBNSGA-II layout as the number of sensors increased from 11 to 13, the spatial configurations of the layouts with 5, 11, and 13 sensors were examined, as shown in Figs. 5(c)–5(k). As the number of sensors increased, IPD-GA retained a dispersed network-wide layout, whereas the locations selected by GSP became increasingly concentrated around

the topological center of the network. In contrast, TBNSGA-II exhibited pronounced spatial clustering, with multiple sensors concentrated in adjacent areas, when the number of sensors increased from 11 to 13. The spatial distribution of reconstruction errors at unmonitored nodes indicated that this substantial clustering reduced the effective perceptual coverage of the sensor network. The different layout patterns may be attributed to differences in the underlying mechanisms of the sensor-

placement strategies. The GSP-based approach determines sensor locations by maximizing node discriminability, which is grounded in the smoothness and spectral properties of graph signals. Specifically, it employs heat kernel diffusion and Laplacian spectral decomposition to evaluate the frequency-domain separability of nodes. However, heat-kernel diffusion relies on the undirected graph Laplacian and inherently exhibits symmetric propagation, the method tends to favor nodes located near the topological center or in regions with densely spaced manholes. This central tendency arises from the uniformity of diffusion across all directions, regardless of actual flow dynamics. Crucially, spatial proximity between nodes does not necessarily correspond to consistent hydraulic flow paths. For instance, a single node may connect to two pipelines with opposing flow directions. As a result, sensor networks designed by GSP perform poorly in reconstruction accuracy compared with IPD-GA. The objective function of TBNSGA-II is formulated based on a trade-off between upstream coverage and search cost. The core assumption underlying this strategy is that the hydraulic states of upstream nodes can be inferred from observations at downstream sensors, and consequently, when the number of available sensors is limited, the algorithm tends to prioritize nodes along the downstream major pipes near the network outfalls in order to maximize upstream coverage. As the number of sensors increases, the algorithm shifts towards a more balanced spatial distribution by optimizing both coverage and search cost. Under this condition, nodes within regions that already exhibit high upstream coverage may continue to be selected, even if neighboring nodes have already been assigned as sensor locations. This leads to deployment redundancy, as multiple sensors are concentrated in specific areas while contributing only marginal improvements to system state reconstruction accuracy.

In contrast, IPD-GA integrates upstream and downstream hydraulic interaction pathways with the GNN message-passing mechanism, explicitly incorporating flow directionality and neighborhood range, thereby producing sensor deployment patterns that are more conducive to the reconstruction of unmonitored node states across different sensor deployment scales.

To examine whether the advantage of IPD-GA over GSP and

TBNSGA-II resulted solely from the matched setting of $k=10$ and $L=10$, the IPD parameter was fixed at $k=10$, and the reconstruction performance of the resulting sensor layouts was evaluated using Edge-STGCN models with varying graph convolution depths ($L=5, 10, 15$, and 20).

As shown in Fig. 6, IPD-GA consistently outperforms GSP and TBNSGA-II across Edge-STGCN models with different graph convolution depths. For $L=5$, the three placement strategies exhibited broadly comparable reconstruction performance. The advantages of IPD-GA over GSP and TBNSGA-II in terms of the area under the reconstruction-performance curve were only 1.7% and 2.1%, respectively. When $L=10$, these advantages increased substantially to 15.8% and 17.4%. For $L=15$, the corresponding advantages were 12.7% and 13.4%, while for $L=20$ they were 9.2% and 6.8%, respectively. These results indicate that although the advantage of IPD-GA was most pronounced when $k=L=10$, its superior performance cannot be solely attributed to the exact correspondence between the IPD scale and a 10-layer message-passing process. Improving the reconstruction capability of the reconstructor benefits all sensor placement strategies rather than only the IPD-based method. For example, when only five sensors were deployed, the reconstruction accuracies of IPD-GA, GSP, and TBNSGA-II were 14.5%, 11.8%, and 12.6%, respectively, when $L=5$. As L increased to 20, these values increased to 76.3%, 67.3%, and 63.3%, respectively.

However, the extent of the observed advantage of IPD-GA appears to be influenced by the reconstruction performance of the reconstructor. The sensor network designed by IPD-GA, GSP, and TBNSGA-II were additionally assessed using a matrix-completion method (supplementary material Part H). Compared with the more pronounced differences observed when Edge-STGCN ($L=10, 15, 20$) was used as the reconstructor, the matrix-completion results showed relatively small differences among the three placement strategies, with substantial overlap within the standard-deviation ranges for most sensor numbers. A similar pattern was observed when the Edge-STGCN ($L=5$) was used as the reconstructor. These findings suggest that a reconstructor with limited reconstruction capability may not fully distinguish the differences among sensor placement strategies in terms of hydraulic-state reconstruction performance, thereby influencing the observed magnitude of

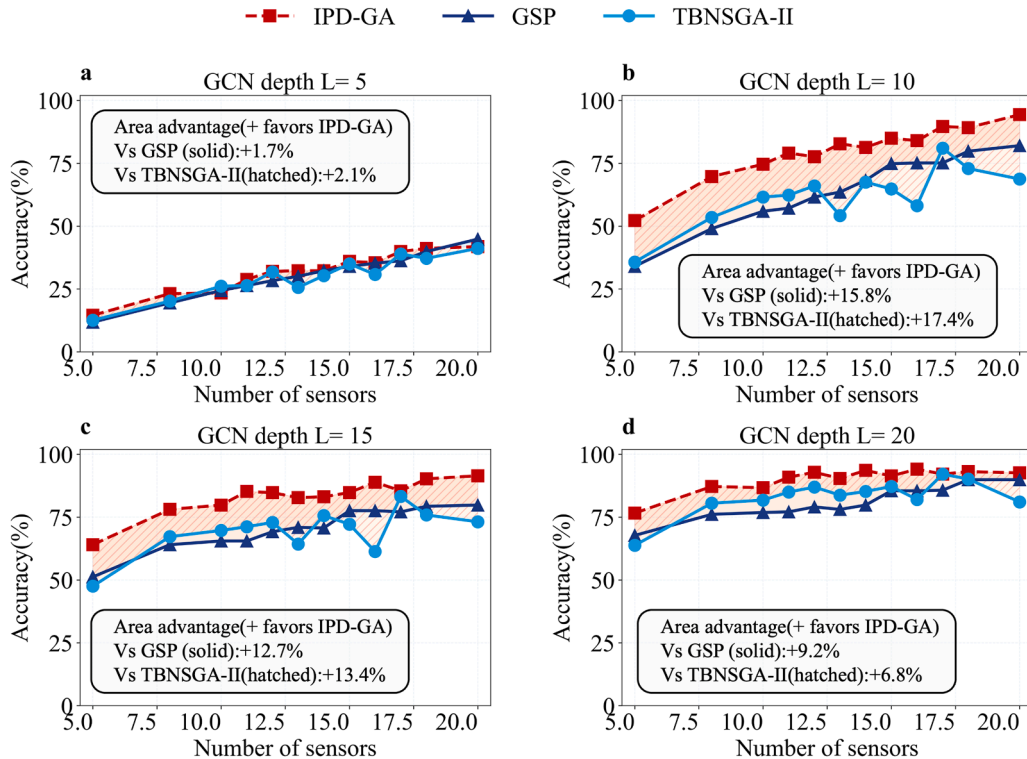


Fig. 6. Effect of GCN depth on the reconstruction performance of sensor placement strategies based on the Edge-STGCN model.

their performance advantages.

4.3. Translating perceptual objectives into effective placement strategies

Although the optimization objectives and performance-evaluation criteria were clearly defined, differences in the layout-generation mechanisms of the optimization frameworks could still affect the performance and engineering applicability of the resulting sensor networks. Under the unified optimization objective of the IPD, this section further evaluated the differences in system-wide reconstruction performance among sensor networks designed using a greedy algorithm, an integer linear programming (ILP) approach, and a genetic algorithm (GA).

As shown in Fig. 7(a), under the representative rainfall event 1#, compared with the ILP and GA optimization algorithms, the sensor

layouts generated by the Greedy algorithm exhibited consistently stronger agreement between the IPD and the APD across different sensor deployment scales. The repeated experiments using the remaining random seeds, provided in the supplementary material (Part I), exhibited the same trend. For example, the ILP achieved 100% coverage of the IPD using only 15 sensors, whereas the GA and Greedy approaches required 23 and 24 sensors, respectively. However, evaluation of the APD revealed that the 15 sensors selected by ILP achieved only 92.10% coverage, representing a 7.90% decrease from the ideal value and yielding an IPD–APD consistency coefficient (η) of 0.92. In comparison, under the same sensor number, the Greedy achieved an APD coverage of 91.35%, 3.2% lower than its ideal value, yielding an η of 0.97, whereas the GA achieved an APD coverage of 88.41%, with a deviation of 4.66% and an η of 0.95. Additional validation using the remaining rainfall

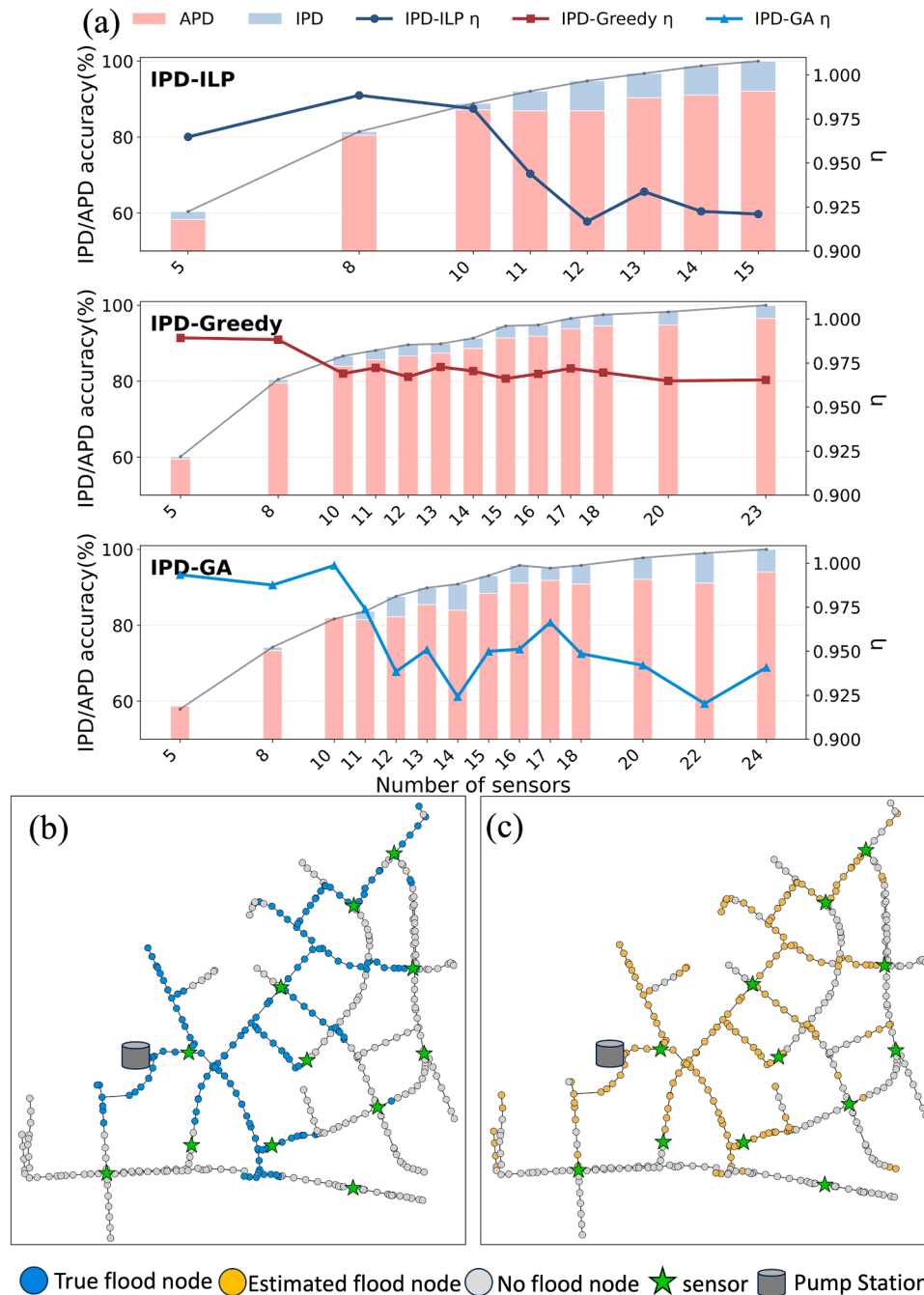


Fig. 7. (a) The variations of IPD, APD, and IPD-APD consistency coefficient η with sensor number. (b-c) The spatial distribution of actual and estimated flooding locations.

events showed that the same pattern remained stable across multiple rainfall scenarios and was particularly evident during more intense storm events, as shown in Fig. S7.

From the perspective of the prescribed IPD coverage objective, ILP was more efficient than Greedy because it achieved complete IPD coverage with fewer sensors. However, the different levels of η exhibited by the three algorithms revealed that the a priori topology-based IPD and the a posteriori reconstruction-based APD were related but not directly equivalent. High IPD coverage did not necessarily guarantee a corresponding level of APD coverage, because topological reachability alone could not fully determine reconstructability. For example, when the number of sensors selected by ILP increased from 10 to 12, the IPD increased, but the APD did not increase correspondingly, causing the consistency coefficient η to decrease. The GA-based layouts were less stable because of random initialization, multiple feasible solutions, and a limited number of iterations, resulting in fluctuations in η . In contrast, Greedy maintained relatively high and stable η values across different sensor numbers, suggesting that its nested layouts and moderate information redundancy may benefit reconstruction performance and robustness. This consistency helped ensure that the sensor numbers and locations determined from the IPD more closely reflected the level of perceptibility that the system could actually achieve, while providing a more robust basis for subsequent sensor-network expansion and layout adjustment. Together with the findings in Section 4.2, these results suggest the potential existence of an optimal level of information redundancy, warranting further investigation in future studies.

From an engineering application perspective, this consistency also held significant practical value. In existing research and engineering practice, sensor placement was typically approached by predefining the number of sensors and then solving for the optimal layout. However, the number of sensors was often selected without a clear theoretical basis and was commonly interpreted under the simplistic assumption that “more sensors lead to better monitoring.” This frequently resulted in increased deployment costs with only marginal improvements in monitoring effectiveness. The consistency between the ideal and practical perceptual domains provided an interpretable and system-level basis for sensor placement decisions. In this study, the Greedy optimization results indicated that approximately 12 sensors were sufficient to achieve nearly 90% IPD coverage ($APD = 86.7\%$, $\eta = 0.96$), whereas increasing the coverage to 100% required 23 sensors. This meant that an additional investment of 11 sensors yielded only a 10% increase in perceptual domain coverage. This result indicated that as the number of sensors increased, the system’s perceptibility exhibited a clear pattern of diminishing marginal returns, and a perceptual saturation point could be quantitatively identified. At the engineering decision-making level, the perceptual saturation point provided an interpretable and quantitative basis for determining the number of sensors. It supported a shift in sensor placement strategy from the traditional “preset quantity followed by layout optimization” paradigm to a demand-driven “deploy-as-needed” approach. This enabled near-optimal monitoring performance with the minimal number of sensors required to meet system-level reconstruction accuracy targets, thereby avoiding unnecessary resource investment.

To further evaluate the practical utility of the 12-sensor network designed by the IPD-based Greedy algorithm, system-wide water levels were reconstructed using the Edge-STGCN model with sensor observations as input under the representative rainfall event 1# and subsequently used to identify potential flooding nodes. As shown in Figs. 7(b) and 7(c), the proposed framework achieved a precision of 0.95, a recall of 0.91, and an FPR of 0.06. These results indicate that the IPD-driven Greedy sensor placement strategy, combined with the reconstruction model, can effectively support flooding-node identification and localization, demonstrating its applicability to urban flood risk monitoring.

Overall, this case study further validated the consistency and effectiveness of the proposed sensor placement framework within an end-to-end monitoring workflow, from sensor deployment optimization to

system state reconstruction, and ultimately to risk-related task applications. By explicitly incorporating system perceptibility into the placement stage, the framework not only improved reconstruction accuracy but also demonstrated robust risk identification performance under real rainfall conditions. The resulting sensor configuration provided a reliable data foundation for risk-aware monitoring and decision support, enabling targeted management and engineering intervention strategies in high-risk areas. This, in turn, enhanced the resilience of urban drainage systems in response to extreme rainfall events.

5. Framework scalability and transferability on another independent drainage network

The proposed framework was further applied to a larger independent drainage network consisting of 660 nodes and 680 pipes, covering an area of 565 ha, with detailed topological information provided in Fig. S8. Based on the findings of Section 4.3, an IPD of 90% was adopted as the target for sensor placement optimization, and the Greedy algorithm was employed to determine the optimal sensor locations and the required number of sensors for this external case study. Under the representative rainfall event 1#, the performance of the designed sensor network in system-wide state reconstruction and flooding-node identification was evaluated, with the results presented in Fig. 8.

The results showed that the framework could determine the required number of sensors according to the prescribed IPD coverage requirement and achieved an APD close to the target IPD coverage. When the reconstructed system-wide node water levels were used to identify flooding nodes, the precision, recall, and false-positive rate were 0.91, 0.74, and 0.11, respectively. The recall of 0.74 indicated that approximately 26% of the actual flooding nodes were not identified. Further analysis of the spatial distribution and topological characteristics of the missed nodes showed that they were mainly located in two types of areas: (1) areas with relatively large topological distances from the deployed sensors and (2) local areas situated on pipe branches or characterized by pronounced changes in pipe invert elevation or drop connections. These characteristics may hinder effective information propagation from monitored nodes to these locations or introduce greater complexity into their local hydraulic responses, thereby causing the reconstructed water levels to underestimate the actual flood levels. These results provided preliminary evidence that the proposed framework could be applied to another drainage-network configuration. However, the reduced recall also indicated that broader validation across drainage networks with different scales, topologies, hydraulic characteristics, flooding patterns, and monitoring conditions was still required before general transferability could be established.

6. Conclusion

Sensor networks serve as critical infrastructure for monitoring the operational states of drainage systems. Under constrained sensing resources, a major challenge for utility operators is to rapidly and rationally allocate sensor locations based on system-level monitoring requirements while achieving the highest possible level of network-wide observability through the designed sensor network. To address this challenge, this study proposed a topology-first and demand-driven end-to-end framework spanning sensor placement, perceptibility assessment, and monitoring applications. The main conclusions are as follows:

- (1) Compared with existing sensor network performance evaluation approaches, the reconstruction model proposed in this study demonstrates significant advantages in capturing spatial dependencies and improving reconstruction accuracy. It establishes a new paradigm for sensor network performance assessment that integrates high precision with physical interpretability.
- (2) The proposed IPD-driven sensor placement strategy consistently achieved higher system-wide state reconstruction accuracy

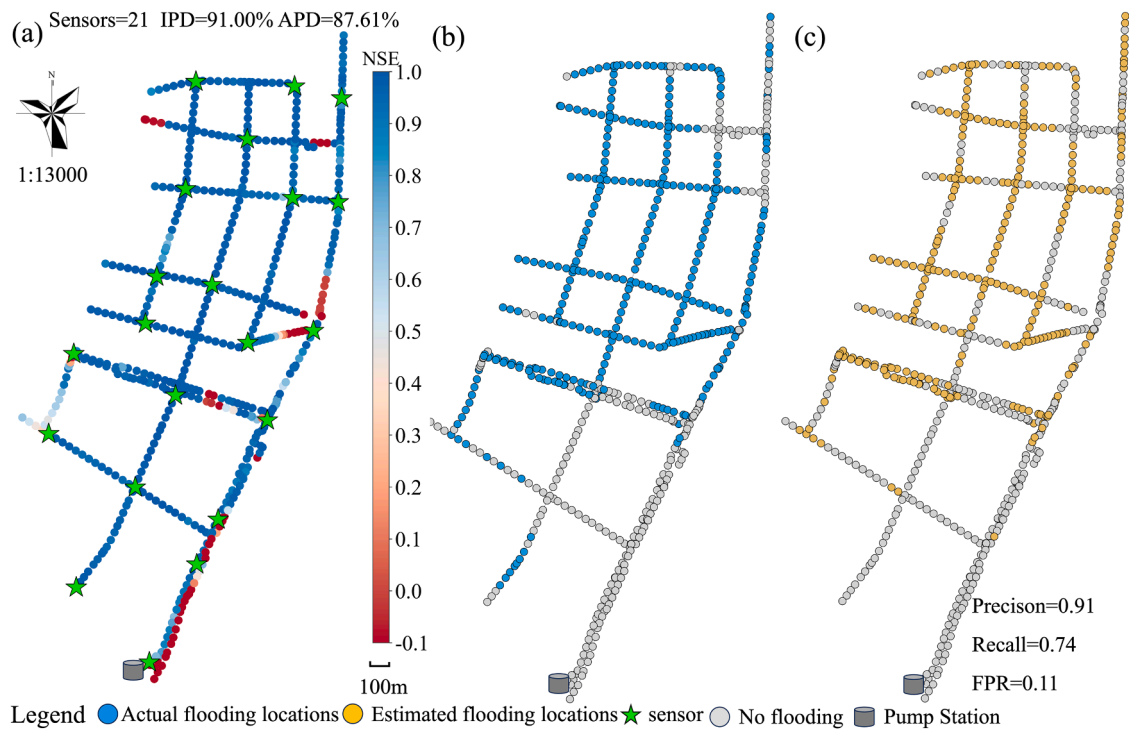


Fig. 8. (a) The spatial distribution of system-wide node water-level reconstructed errors. (b-c) The spatial distribution of actual and estimated flooding locations.

across different sensor deployment scales, demonstrating its effectiveness in improving network perceptibility.

- (3) The high IPD–APD consistency achieved by the Greedy-based placement strategy enables the effective perceptual range of the designed sensor network to be estimated a priori, providing a theoretical basis for transitioning sensor deployment from pre-defined configurations to demand-driven design.

Overall, the proposed framework provides a simple, practical, and effective closed-loop solution that integrates sensor deployment, system-state perception, and risk-aware monitoring, thereby supporting decision-making in data-limited urban drainage systems. Nevertheless, several limitations of the proposed framework warrant further investigation. Future work should focus on (1) establishing the underlying mechanism of information redundancy trade-offs and elucidating their impacts on reconstruction performance and robustness, (2) incorporating field measurements of deployment costs, boundary conditions, and operational processes to further assess the engineering applicability of the proposed framework, and (3) exploring the potential of reconstructed hydraulic states to support numerical model calibration and digital twin updating.

Data availability

The proposed models are publicly available at <https://github.com/holylove9412/End-to-End-Framework>.

CRediT authorship contribution statement

Li He: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Jun Nan:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision. **Lei Chen:** Methodology, Software, Validation, Writing – review & editing. **Xuesong Ye:** Formal analysis, Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2026.126713](https://doi.org/10.1016/j.watres.2026.126713).

Reference

- Bartos, M., Kerkez, B., 2021. Observability-Based Sensor Placement Improves Contaminant Tracing in River Networks. *Water Resources Research* 57 (7), e2020WR029551. <https://doi.org/10.1029/2020WR029551>.
- Cheng, M., Li, J., 2023. Optimal sensor placement for leak location in water distribution networks: A feature selection method combined with graph signal processing. *Water Research* 242, 120313. <https://doi.org/10.1016/j.watres.2023.120313>.
- Crowley, G., Tait, S., Panoutsos, G., Speight, V., Esnaola, I., 2025. Information-theoretic sensor placement for large sewer networks. *Water Research* 268, 122718. <https://doi.org/10.1016/j.watres.2024.122718>.
- Garzón, A., Kapelan, Z., Langeveld, J., Taormina, R., 2024. Transferable and data efficient metamodeling of storm water system nodal depths using auto-regressive graph neural networks. *Water Research* 266, 122396. <https://doi.org/10.1016/j.watres.2024.122396>.
- Giudicianni, C., Herrera, M., Di Nardo, A., Greco, R., Creaco, E., Scala, A., 2020. Topological Placement of Quality Sensors in Water-Distribution Networks without the Recourse to Hydraulic Modeling. *Journal of Water Resources Planning and Management* 146 (6), 04020030. [https://doi.org/10.1061/\(ASCE\)JWR.1943-5452.0001210](https://doi.org/10.1061/(ASCE)JWR.1943-5452.0001210).
- Guo, K., Guan, M., Yan, H., Xia, X., 2023. A spatially distributed hydrodynamic model framework for urban flood hydrological and hydraulic processes involving drainage

- flow quantification. *Journal of Hydrology* 625, 130135. <https://doi.org/10.1016/j.jhydrol.2023.130135>.
- He, L., Nan, J., Chen, L., Ye, X., Ji, S., Chen, Z., Zhang, Y., Wu, F., Liu, B., Ge, Z., Che, Y., 2025a. Spatiotemporal graph convolutional network using sparse monitoring data for accurate water-level reconstruction in urban drainage systems. *Journal of Hydrology* 652, 132681. <https://doi.org/10.1016/j.jhydrol.2025.132681>.
- He, L., Nan, J., Ye, X., Chen, L., Ji, S., Chen, Z., Xiao, Q., 2025b. A graph neural network using physical attributes to improve the system-wide nodal water-level prediction in sparsely monitored urban drainage systems. *Journal of Hydrology* 663, 134306. <https://doi.org/10.1016/j.jhydrol.2025.134306>.
- Khorshidi, M.S., Nikoo, M.R., Sadegh, M., 2018. Optimal and objective placement of sensors in water distribution systems using information theory. *Water Research* 143, 218–228. <https://doi.org/10.1016/j.watres.2018.06.050>.
- Khorshidi, M.S., Nikoo, M.R., Taravatrouy, N., Sadegh, M., Al-Wardy, M., Al-Rawas, G. A., 2020. Pressure sensor placement in water distribution networks for leak detection using a hybrid information-entropy approach. *Information Sciences* 516, 56–71. <https://doi.org/10.1016/j.ins.2019.12.043>.
- Kim, K., Ban, M.J., Kim, S., Park, M.-H., Stenstrom, M.K., Kang, J.-H., 2022. Optimal allocation and operation of sewer monitoring sites for wastewater-based disease surveillance: A methodological proposal. *Journal of Environmental Management* 320, 115806. <https://doi.org/10.1016/j.jenvman.2022.115806>.
- Li, Z., Liu, H., Zhang, C., Fu, G., 2024. Real-time water quality prediction in water distribution networks using graph neural networks with sparse monitoring data. *Water Research* 250, 121018. <https://doi.org/10.1016/j.watres.2023.121018> (1).
- Liu, W., Gao, W., Sun, Y., Xu, M., 2008. Optimal sensor placement for spatial lattice structure based on genetic algorithms. *Journal of Sound and Vibration* 317 (1–2), 175–189. <https://doi.org/10.1016/j.jsv.2008.03.026>.
- Marlim, M.S., Kang, D., 2021. Optimal Water Quality Sensor Placement by Accounting for Possible Contamination Events in Water Distribution Networks. *Water* 13 (15), 1999. <https://doi.org/10.3390/w13151999>.
- Oh, J., Bartos, M., 2025. Scalable, adaptive and risk-informed design of hydrological sensor networks. *Nature Water* 3 (10), 1144–1154. <https://doi.org/10.1038/s44221-025-00496-7>.
- Pang, S., Wang, X., Melching, C.S., Feger, K.-H., 2020. Development and testing of a modified SWAT model based on slope condition and precipitation intensity. *Journal of Hydrology* 588, 125098. <https://doi.org/10.1016/j.jhydrol.2020.125098>.
- Romero-Ben, L., Alves, D., Blesa, J., Cembrano, G., Puig, V., Duviella, E., 2023. Leak detection and localization in water distribution networks: Review and perspective. *Annual Reviews in Control* 55, 392–419. <https://doi.org/10.1016/j.arcontrol.2023.03.012>.
- Taha, A.F., Wang, S., Guo, Y., Summers, T.H., Gatsis, N., Giacomoni, M.H., Abokifa, A.A., 2021. Revisiting the Water Quality Sensor Placement Problem: Optimizing Network Observability and State Estimation Metrics. *Journal of Water Resources Planning and Management* 147 (7), 04021040. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001374](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001374).
- Tobin, G.A., 1989. *Applied Hydrology*, McGraw-Hill Series in Water Resources and Environmental Engineering. Eos, Transactions American Geophysical Union 70 (10), 149. <https://doi.org/10.1029/89EO00083>. –149.
- Tornyeviadzi, H.M., Owusu-Ansah, E., Mohammed, H., Seidu, R., 2024. Node search space reduction for optimal placement of pressure sensors in water distribution networks for leakage detection. *Alexandria Engineering Journal* 94, 325–338. <https://doi.org/10.1016/j.aej.2024.03.037>.
- Wang, H., Meng, Y., Xu, H., Wang, H., Guan, X., Liu, Y., Liu, M., Wu, Z., 2024. Prediction of flood risk levels of urban flooded points though using machine learning with unbalanced data. *Journal of Hydrology* 630, 130742. <https://doi.org/10.1016/j.jhydrol.2024.130742>.
- Wang, S., Xu, K., Zhou, Y., 2024. Cost-effective sensor placement optimization for large-scale urban sewage surveillance. *Sustainable Cities and Society* 103, 105250. <https://doi.org/10.1016/j.scs.2024.105250>.
- Wang, S., Zhang, X., Wang, J., Tao, T., Xin, K., Yan, H., Li, S., 2023. Optimal sensor placement for the routine monitoring of urban drainage systems: A re-clustering method. *Journal of Environmental Management* 335, 117579. <https://doi.org/10.1016/j.jenvman.2023.117579>.
- Wang, W., Wang, D., Singh, V.P., Wang, Y., Wu, J., Wang, L., Zou, X., Liu, J., Zou, Y., He, R., 2018. Optimization of rainfall networks using information entropy and temporal variability analysis. *Journal of Hydrology* 559, 136–155. <https://doi.org/10.1016/j.jhydrol.2018.02.010>.
- Winter, C.D., Palleti, V.R., Worm, D., Kooij, R., 2019. Optimal placement of imperfect water quality sensors in water distribution networks. *Computers & Chemical Engineering* 121, 200–211. <https://doi.org/10.1016/j.compchemeng.2018.10.021>.
- Wu, Y., Zheng, F., Yang, Y., Zhang, K., Du, K., Duan, H., Savic, D., Kapelan, Z., 2025. Enhancing inflow and infiltration detection in urban sewer networks with a new deterministic sensor placement method. *Water Research* 268, 122591. <https://doi.org/10.1016/j.watres.2024.122591>.
- Zhang, Z., 2024. Graph neural network-based surrogate modelling for real-time hydraulic prediction of urban drainage networks. *Water Research*. <https://doi.org/10.1016/j.watres.2024.122142>.
- Zheng, Y., Jin, X., Wei, J., Zhou, Y., Zhang, Y., 2025. A novel framework for optimization and evaluation of sensors network in urban drainage system. *Water Research* 270, 122833. <https://doi.org/10.1016/j.watres.2024.122833>.